

Experiment 3: RTL Simulation Report

FPGA ECG FIR Filtering and Peak Analysis | PYNQ-Z2 / XC7Z020

Result: PASS - 28,672/28,672 exact Q4.12 samples; maximum absolute error 0 LSB.

This is a genuine independent digital simulation of the unchanged HLS-generated FIR core with Icarus Verilog 12.0. It is separate from the original HLS C simulation and the saved physical-board demonstration. The Zynq processing system, DDR and vendor AXI DMA are outside this unit test.

Method and acceptance checks

The clock period is 10 ns. The core count input remains 2048 and ap_start is held high while transactions run. A deterministic driver inserts input gaps; a deterministic sink inserts three consecutive not-ready cycles in every thirteen. Checks require exact sample values, stable output data/sidebands while stalled, KEEP/STRB preservation, one TLAST per batch, and completion of every invocation.

Case	Payload samples	Zero flush	Batches	Observed errors
Impulse	2048	2048	2	0
5 Hz + 50 Hz sine	2048	2048	2	0
Signed random / extrema	4096	2048	3	0
Recorded ECG	12288	2048	7	0

Total: 20,480 payload samples plus 8,192 checked zero-flush samples = 28,672 samples over 14 batches. Random stimuli use seed 20261009 and include signed extrema. Seven ECG batches include one flush batch and six data batches.

Reference and reset behavior

An independent integer Python model uses each signed Q4.12 product shifted right by four bits, a signed 24-bit accumulator wrap, an eight-bit final shift and signed 16-bit output wrap. Its ECG reference was additionally checked against the original saved board CSV. Every scenario starts with the same 2048-zero history flush used by the hardware notebook. Control reset is not treated as clearing the generated static FIR delay state. The testbench waits for ap_done after the final output before changing scenarios.

Reproduce: run python Testbench/run_rtl_sim.py with NumPy, iverilog and vvp installed. Actual commands, simulator outputs, return codes and source checksums are preserved in transcript.txt and rtl_run_metadata.json. All generated DUT RTL remains unchanged.

Actual RTL waveform

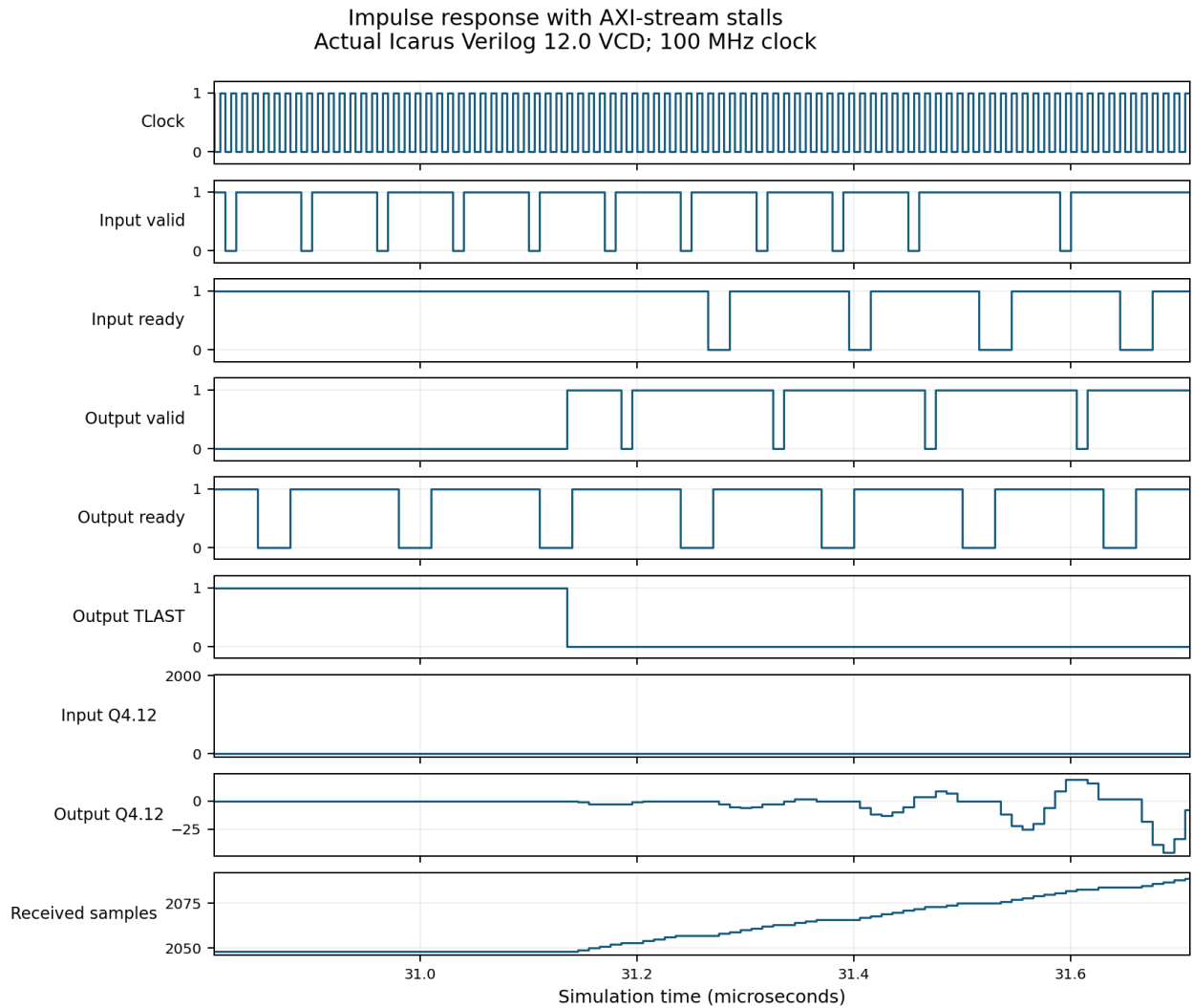


Figure 1. Signals decoded from `rtl_waveform.vcd`. The output is the generated RTL response to the checked impulse input; the not-ready intervals exercise AXI-stream backpressure.

Full evidence includes `waveform_tlast.png`, `rtl_waveform.vcd.gz` and `rtl_output_samples.csv`. The CSV has one row per accepted output and records actual/expected signed values. The compressed VCD preserves the entire run, not just this plotted window.

Scope limitation: a unit RTL pass establishes the tested core behavior and stream protocol under these stimuli. It does not replace the existing Vivado timing/DRC reports or prove complete PS/DMA integration in simulation.